

Richard MEDINA RODRIGUEZ

PhD in Applied Mathematics

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Apply mathematics PhD specializing in PDEs, the long-time behavior of kinetic equations, hydrodynamic limits, and boundary effects. Experienced in scientific programming with Python and C#. Interested in the analysis and numerical simulation of particle systems, with emphasis on well-posedness and asymptotic limits.

Professional Experience

2025 - Present	Research and Teaching Fellow (A.T.E.R.), Université Paris Dauphine – PSL, France - Well-posedness and long-time behavior of a non-isothermal Boltzmann equation in the regime of the hydrodynamic limit - Working on the analysis and simulation of coupled fluid-(stochastic)-particles system PythonGit$\LaTeX$
2022 - 2025	PhD in Applied Mathematics, Université Paris Dauphine – PSL, France - Effects of heat thermostats on the well-posedness and long-time behavior of kinetic Fokker–Planck equations - Boundary effects for the Boltzmann equation in cylindrical domains PythonMathematicaMATLAB$\LaTeX$
2022	Master Internship, Université Paris Dauphine – PSL, France Developed hypocoercivity methods for linearized kinetic operators in cylindrical domains $\LaTeX$
2020 - 2021	Junior Assistant Professor, Universidad de La Habana, Cuba - Analysis of bifurcations and long-time convergence of dynamical systems - Worked on the improvement of a spline-based software for geometric design - Implementation of an interpolation method to approximate curves PythonC#Mathematica$\LaTeX$

Publications

2026	The Boltzmann equation with non-isothermal Maxwell boundary conditions.
2025	The Boltzmann equation on smooth and cylindrical domains with Maxwell boundary conditions.
2025	Existence and Stability of Non-Equilibrium Steady States of a Weakly Non-Linear Kinetic Fokker-Planck Equation in a Domain. J Stat Phys 193, 47. Joint work with J. Evans.
2024	Constructive Krein-Rutman result for Kinetic Fokker-Planck equations in a domain. Joint work with K. Carrapatoso, P. Gabriel, and S. Mischler.

Education

2022 - 2025	PhD in Applied Mathematics , CEREMADE, Université Paris Dauphine - PSL, France.
2021 - 2022	Master in Applied and Theoretical Mathematics , Université Paris Dauphine - PSL, France. <i>With Honors.</i>
2016 - 2020	Bachelor in Mathematics , Universidad de La Habana, Cuba. <i>Summa Cum Laude.</i>

Awards & Grants

2022	MathInParis2020 Fellowship. PhD grant awarded by the Fondation de Sciences Mathématiques de Paris.
2021	PSL PhD Track Grant 2021. Master's Excellence Scholarship awarded by the Université PSL.
2020	Prize for Scientific Merit awarded by the Universidad de La Habana.

Selected Presentations

Feb. 2026	The Boltzmann equation with non-isothermal Maxwell boundary conditions , presented at the SPIKE Seminar, Institut Henri Poincaré (IHP), France.
Jan. 2025	Brief introduction to the Boltzmann equation , presented at the Young Researchers Seminar, Université Paris Dauphine - PSL, France.
Nov. 2024	Constructive Krein-Rutman result for kinetic Fokker-Planck equations in a domain (joint work with P. Gabriel, K. Carrapatoso and S. Mischler), presented at the Junior Analysis Seminar, Imperial College London, United Kingdom.
Oct. 2024	The Boltzmann equation on C^1 and cylindrical domains near the hydrodynamic limit , presented at the Seminar Partial Differential Equations and their Applications, University of Warwick, United Kingdom.
Jun. 2024	The Boltzmann equation in a cylinder near the hydrodynamic limit , presented at the conference “Analysis of PDEs in Mathematical Physics”, University of Bath, United Kingdom.
Feb. 2023	Weighted L^∞ estimates for the Boltzmann equation , presented at the MAFRAN Days, King's College, United Kingdom.

Organization of Seminars and Conferences

- Jun. 2024 Co-organizer of the **Young Researchers Days** conference, held at *Domaine de la Tour*, Saint-Pierre-Canivet, France.
- 2023 – 2024 Co-organizer of the **Young Researchers Seminar** at the Université Paris Dauphine - PSL.

Teaching Experience

- 2024 - 2026 Practical lessons of **Analysis I, II, and Algebra IV & Numerical Methods** at *Paris Dauphine* (Mathematics program).
- 2023 Practical lessons of **Mathematical Methods** at *Paris Dauphine* (Economics program).
- 2020 - 2021 Practical lessons of **Complex Analysis** at *Universidad de La Habana* (Mathematics program).

Outreach

- May 2024 Volunteer animator at the **Salon de Culture et Jeux Mathématiques** at Place Saint-Sulpice, Paris, France.
- March 2024 Volunteer judge at the **Math en Jeans 2024 Congress** at the Université Paris Dauphine - PSL, France.

Languages

French (C1), English (TOEIC C1, 2022), Spanish (Native).

Interests

Cycling, Running, Camping, Scientific Vulgarisation.

Referees

Prof. Stéphane Mischler

*CEREMADE, Université Paris Dauphine - PSL;
& Institut Universitaire de France (IUF).*

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Prof. Josephine Evans

*Warwick Mathematics Institute,
University of Warwick.*

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Prof. Kleber Carrapatoso

*Centre de Mathématiques Laurent Schwartz (CMLS),
École Polytechnique.*

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Prof. Pierre Gabriel

*Institut Denis Poisson,
Université de Tours.*

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